

SMITHIAN FOLD THEORY - FOUNDATION BRANCH PAPER 001

From Nothing to Fold

A Premise-Free, Parameter-Free and Machine-Closed Foundation for
Smithian Fold Theory

Ernos Labs

Open Source Science Platform and Knowledge Tree

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Third clean-room reconstruction - Foundation current-knowledge inventory complete

Ten admitted derivations - 2,450 generated candidate classes

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Paper: CC BY 4.0 - Code: Apache-2.0

Abstract

This paper reports the completed foundation branch of the third clean-room reconstruction of Smithian Fold Theory (SFT). The reconstruction begins with one premise-free operational theorem - *there is no nothing* - and derives, in dependency order, the structural One, exact positive finite count, exact positive part coordinates, cross-partition part equivalence, the minimal Fold, complete finite Fold-word support, the recursive foundational form grammar, canonical form identity and the one-way boundary from exact derivation to conventional measurement. No conventional number axioms, numerical zero, negative quantity, irrational or imaginary proof value, floating-point proof, completed infinity, fitted parameter, trained weight, target answer or earlier SFT result is admitted as a premise.

Each result is processed by one fail-closed admission engine. Admission requires source-bound registration, a generated complete candidate grammar, one decision per candidate, exactly one survivor, minimality, named-shape uniqueness, an exact boundary, four adverse-control classes, cryptographic sealing and implementation-distinct recomputation. The ten claims execute 2,450 generated candidate classes. Each claim has depth-independent closure and an independently replicated engine receipt. The complete repository verification executes 96 unit, adverse-control and end-to-end tests, observes all 1,264 executable lines of the 15-module core engine, and freshly recomputes all ten claims in order.

The result is deliberately foundational. It does not claim the later branches of mathematics, information, computation, physics, chemistry or biology. It establishes the exact objects and evidence boundary those branches must cite. Every candidate, rejection, proof identity, control, certificate and receipt is published as machine-readable evidence alongside this paper.

1. Scientific claim and exact boundary

The claim of this paper is not that a repository exists, that tests pass, or that familiar ideas can be rewritten in new vocabulary. The claim is narrower and mechanically inspectable:

<i>Within the frozen SFT V3 foundation current-knowledge inventory, all ten</i>
<i>registered obligations have depth-independent, model-admitted and</i>
<i>independently replicated receipts, and no registered foundation obligation</i>
<i>remains unclassified or open.</i>

The frozen inventory hash is

sha256:ebf5f092065bbcc1d6324514820d9111b05f374651dc646f1efb8e240945420. The inventory covers the operational root, structural identity, positive finite count, lawful parts, part equivalence, the first Fold, finite Fold composition, finite form generation, form identity and the derivation-to-measurement interface. It does not include arithmetic operations, probability, computation, physical dynamics or application translations. Those are downstream branches.

No application was allowed to select these laws. Fold Protein, Fold Chess, Fold Go and Unison AI remain outside the V3 derivational authority. No empirical data is needed to establish these formal structural results. The final foundation law specifies how later empirical branches must interact with external data.

2. Constitution of the derivational domain

The operational root is the sole root theorem. Every later dependency must end at that identity through model-admitted receipts. Registered derivations contain empty axiom and free-parameter lists. The permitted mathematical domain consists of the structural One, generated positive finite traces, exact positive parts of the One, exact held labels, fibres, words, relations and finite forms.

The derivational domain excludes semantic numerical zero, negative quantities, irrational and imaginary values, floating-point proof quantities, completed infinity and an ungenerated continuum. The empty One word is a structural base form, not numerical zero. Opposed identities are represented by distinct held labels rather than signed magnitudes.

Python necessarily uses host indices, empty containers, Boolean flags and process return codes. These mechanics implement checks but possess no SFT mathematical authority. Exact proof values are guarded separately. Conventional decimal measurements may exist only after the one-way measurement boundary and cannot flow backward into law selection.

3. What “forced” means here

Forcing is not satisfied by naming plausible ingredients. A result is admitted only after the complete declared same-grammar candidate domain has been generated and executed. A candidate generator states its dimensions and exact boundary. Every generated candidate has a stable identifier and trace hash. Every candidate receives a survival decision, reason and proof hash. Exactly one candidate must survive.

The survivor must also pass two distinct structural tests. Minimality rejects a form when removing an element preserves the required result or when the form contains an unforced addition. Named-shape uniqueness rejects a form when a different same-boundary structure also satisfies the requirements. A finite enumeration alone is conditional; every claim in this paper also carries a base/successor or structural-induction certificate establishing that its classification does not depend on a maximum execution depth.

Four controls are mandatory: a false premise, changed source, tampered artifact and boundary violation. A separate process with an implementation hash distinct from the derivation source must regenerate the candidate identities and unique result from declared inputs. The engine seals the registration, census, decisions, closure and controls before accepting that external certificate.

4. Dependency chain and execution census

Order	Claim	Generated candidates	Closure	Meaning
1	SFT-ROOT-THERE-IS-NO-NOTHING	2	depth-independent	Operational occurrence cannot be nothing
2	SFT-FOUNDATION-ONE-001	6	depth-independent	Complete addition-free self-whole
3	SFT-FOUNDATION-COUNT-001	10	depth-independent	Complete nonempty finite generation trace
4	SFT-FOUNDATION-PART-001	192	depth-independent	Exact held/whole part coordinate
5	SFT-FOUNDATION-FOLD-001	288	depth-independent	Minimal nontrivial returning distinction
6	SFT-FOUNDATION-PART-EQUIVALENCE-001	1,024	depth-independent	Common-refinement pairing equivalence
7	SFT-FOUNDATION-FOLD-ASSEMBLY-001	192	depth-independent	Complete finite held-label word support
8	SFT-FOUNDATION-FORM-GRAMMAR-001	288	depth-independent	One/Fold recursive productions
9	SFT-FOUNDATION-FORM-ENFORCEMENT-001	192	depth-independent	Canonical construction-trace identity
10	SFT-FOUNDATION-MEASURE-VALUE-BOUNDARY-001	256	depth-independent	One-way sealed measurement interface

The total is 2,450 generated candidate classes. This number is a report of the finite class products executed by the registered generators. It is not a claim that the set of generated finite Fold forms has a completed total cardinality. Depth-independent closure comes from the claim-specific induction certificates.

5. Derivation I: the operational root

5.1 Question

Can an admissible statement, denial, record, proof object or derivational object be nothing?

5.2 Generated alternatives

The root generator distinguishes absence of presentation from presentation. An absent presentation supplies no counterexample object. A presented denial or counterexample is itself an occurrence. These two operational classes exhaust the declared boundary without installing a metaphysical object called nothing.

5.3 Forced result

The survivor is the operational-occurrence class. A presented denial performs or contains denial and therefore instantiates something. Absence cannot present itself as an object that defeats the statement. The theorem has no dependency, premise, axiom or parameter.

5.4 Closure and controls

The classification is independent of the content or depth of a presentation: anything operationally presented occurs; what is not presented supplies no presented counterexample. False-denial, source-drift, artifact and boundary controls pass. A separate implementation regenerates both classes.

Engine receipt: sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669.

6. Derivation II: the structural One

6.1 Question and grammar

The admitted occurrence requires an exact representation before magnitude or arithmetic exists. A representation is classified by root coverage - none, proper or complete - and by whether unforced extra material is absent or present. The generated product has six forms.

6.2 Eliminations

No coverage omits the occurrence. Proper coverage presupposes a cut and loses the complete admitted identity. Any extra-bearing form introduces material not supplied by the only dependency. The unique survivor is `exact-self-whole`: complete coverage without addition.

6.3 Meaning and closure

The One is the complete self-whole of an admitted occurrence, not a conventional numeral or measured magnitude. Removing retained identity makes it incomplete; adding anything makes it nonminimal. The coverage/addition classification is independent of content, notation, size and internal depth.

Engine receipt: sha256:68332624c276dc5d0ddf568a26b95e20d7d5665fcdb7e825ff1b7b19bf1d51c.

7. Derivation III: exact positive finite count

7.1 Why count must be derived

Later partitions require a lawful statement of how many generated occurrences are present. Importing natural-number axioms would violate the clean-room boundary. Count is therefore derived from a registered nonempty finite generation trace.

7.2 Ten-form grammar

Representations are classified by coverage of generated occurrences - none, proper or complete - by whether retained occurrences appear once or at least one is duplicated, and by whether an occurrence outside the trace is added. No coverage has no applicable retained multiplicity, producing two forms. Proper and complete coverage each produce four. The total is ten.

7.3 Survivor

No coverage retains no registered occurrence. Proper coverage omits at least one. Duplication changes the trace, and extra material is ungenerated. Only `complete-coverage__once__no-extra` survives. Exact positive finite count is the

identity of the complete generation trace, not a semantic zero or signed magnitude.

7.4 Depth-independent certificate

The structural One supplies the nonempty base. Appending one generated occurrence makes the previous complete trace a proper trace of the extension; retaining the new terminal once restores completeness. This successor argument repeats at every finite generated depth and never forms a completed infinity.

Engine receipt: sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23.

8. Derivation IV: exact positive part coordinates

8.1 Registered boundary

This theorem does not assume that every proposed partition exists. It asks for the exact coordinate of a nonempty selection within a registered finite partition of the One. A lawful partition must be complete, disjoint and equal-fibred.

8.2 The 192-form product

Seven dimensions are generated: complete or incomplete whole coverage; disjoint or overlapping fibres; equal or unequal fibres; empty, contained or outside selection; held-count presence; whole-count presence; and extra-data presence. Their product contains 192 forms.

Incomplete coverage is not the One. Overlap counts shared structure repeatedly. Unequal fibres prevent counts from naming equal portions without another scale. An empty selection is not positive. An outside selection exceeds the whole. Without the held count the selected portion is unidentified; without the whole count it has no partition reference. Any further scale is unforced.

8.3 Forced coordinate

The unique survivor is `complete__disjoint__equal__within__held-present__whole-present__no-extra`. Thus a lawful part is represented by the exact positive count of held fibres and the exact positive count of the complete equal-fibre partition. Containment forces the held selection to remain through the One.

This result preserves construction identity. It does not yet identify two coordinates generated from different partitions; that question is answered by the next theorem.

Engine receipt: sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728.

9. Derivation V: the minimal structural Fold

9.1 Question

What is the minimal parameter-free nontrivial transformation of the structural One that distinguishes parts while preserving the complete whole and a return?

9.2 The 288-form grammar

The generator crosses whole coverage, fibre overlap, fibre equality, count extension class, label relation, return relation and extra-data presence. Count extension is classified as identity, first extension or later extension. Labels are absent, identical or distinct-held. The complete product contains 288 forms.

9.3 Eliminations and survivor

Incomplete or overlapping structures fail exact whole partition. The unqualified One supplies no selector for unequal allocation, so inequality requires added information. Identity is trivial. The first generated successor is already nontrivial, making later extensions nonminimal. Absent or identical labels fail distinction. Without a return relation the complete assembly is not identified with its source One. Extra data violates parameter freedom.

The unique survivor is `complete__disjoint__equal__first-extension__labels-distinct-held__return-present__no-extra`. The first extension consists of the base occurrence and one generated successor: the forced positive count conventionally read as two. The labels are structural held identities, never positive and negative numerical quantities. This is the minimal Fold.

9.4 Generality

The base/successor count law proves first-extension minimality against every later finite extension. The result is not inferred from testing a fixed maximum arity.

Engine receipt: sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003.

10. Derivation VI: cross-partition part equivalence

10.1 Why conventional reduction is not imported

Held/whole coordinates from different equal partitions may identify the same part. Conventional cross-multiplication, division, decimal comparison and fraction reduction are not premises. Equivalence is reconstructed from generated finite refinement and exact pairing.

10.2 Pair-cell common refinement

For every generated fibre label of partition A, the generator emits one cell with every generated fibre label of partition B. Each cell retains the ordered pair of source labels. Grouping cells by their first label refines A; grouping by their second label refines B. This is nested finite generation, not an imported multiplication law.

The held selection of A lifts to every pair cell descending from a held A fibre. The selection of B is lifted symmetrically. The coordinates identify the same part precisely when their lifted selected traces admit a complete one-to-one and onto pairing.

10.3 The 1,024-form witness grammar

The generator classifies refinement coverage, overlap, fibre equality, refinement of A, refinement of B, complete lifting of both selections, four pairing classes and extra-data presence. Exactly one witness form survives: complete, disjoint, equal, doubly refining, doubly lifted, bijective and addition-free.

10.4 Equivalence laws

An identity pairing establishes reflexivity. Reversing every pair establishes symmetry. Chaining the unique mates of two bijections establishes transitivity. Uniform further refinement appends the same complete descendant construction on both sides and preserves pairing existence. A generated one-of-two versus two-of-four example pairs completely; a one-of-two versus one-of-three control does not.

Engine receipt: sha256:4f7c913f240527ec64d7688bf57fe95ae508f5445786dcbbdc2e6a7094476680.

11. Derivation VII: complete finite Fold-word support

11.1 Recursive support law

The base support contains the structural empty One word. This is a form, not numerical zero. At each generated Fold successor, append each distinct held label once to every prior word. This produces all and only the next support. Distinct predecessors or distinct appended labels have distinct construction traces.

11.2 Representation grammar

The 192 alternatives classify Fold-trace coverage, word-length consistency, support completeness, word duplication, transition records, return records and extra data. Only complete coverage with equal-length unique complete support, all transitions and returns, and no additions survives.

11.3 Induction and finite controls

The base and successor clauses prove the law at every generated finite Fold depth. The independent validator also regenerates the first five nontrivial supports, whose generated counts are 2, 4, 8, 16 and 32. Those values are witness checks, not an imported exponential law and not the generality certificate.

Engine receipt: sha256:8fa632be9e4048a5f2cf6749da4110e3a7b8b4d937051610dcce0583d7ea610d.

12. Derivation VIII: the complete foundational form grammar

12.1 From words to nested forms

Words name paths, but the foundation also requires complete nested assemblies, including nonuniform finite trees. The candidate production grammar classifies base inclusion, child arity, edge-label relation, child provenance, return relation, termination and extra constructors. It contains 288 forms.

12.2 Forced productions

The sole terminal production is the structural One. The sole recursive production is a Fold with exactly two previously generated child forms, one on each distinct held-labelled edge, together with the Fold return relation. Every construction trace terminates finitely in One leaves.

One child omits a forced fibre. Later arity imports a nonminimal branching rule. Absent or same labels erase distinction. Ungenerated children or extra constructors exceed the dependency grammar. Missing return loses whole identity, and missing finite termination admits an uncompleted regress.

Structural induction establishes both directions: every production output is a finite One/Fold form, and every finite assembly from the admitted One and Fold parses through these productions.

Engine receipt: sha256:b768689e7b061fdafa3a620befef79ddcee8bf5750cbbaf19f7b2a3156942be71.

13. Derivation IX: canonical form enforcement

13.1 Exact construction trace

The One maps to its terminal trace. A Fold maps recursively to a Fold marker, the held-a child trace, the held-b child trace and a return marker. Production order is generated by the Fold and is not chosen by lexical sorting.

13.2 The 192-form enforcement grammar

Representations are classified by source coverage, node multiplicity, held-label preservation, child-relation preservation, return preservation, canonical trace status and extra data. Only complete coverage with every node once, all labels, children and returns preserved, canonical recursive order and no addition survives.

Two generated forms are equivalent exactly when their canonical traces agree. This equality is reflexive, symmetric and transitive and preserves rooted, held-labelled structure. Swapping structurally different children changes the trace and is rejected; reconstructing the same trace is accepted.

The base One trace and recursive Fold trace provide depth-independent structural induction. Any omission loses source identity; duplication or addition is nonminimal; changing a relation names another form.

Engine receipt: sha256:84d18de18b1345bf506138493aebd7808bb0bd00cc3b318040e00751929c7c8a.

14. Derivation X: the one-way measurement boundary

14.1 The problem

Later SFT branches must confront conventional external data. Measurement cannot be prohibited, but permitting data to select the law would destroy clean forcing. The boundary must therefore be a derived, executable relation.

14.2 The 256 interface forms

The generator crosses four information-flow classes with proof exactness, target-opening order, row preservation, source identity, law mutation and extra rules. Scientific testing requires a consequence to leave a derivation. Clean forcing forbids targets, measurements and scores from flowing back into law selection. These requirements eliminate disconnected, inward and bidirectional flows.

The unique survivor is `derivation-to-measurement__proof-exact__target-after-seal__rows-complete__source-identified__law-unchanged__no-extra`.

14.3 Operational meaning

The law remains exact and source-bound. Its consequence is sealed before a separately held target is opened. The evaluator records an identified external source and preserves every favorable, unfavorable, failed and tampered row. Conventional decimals remain measured records. Measurement may validate, invalidate or bound the claim but cannot rewrite the sealed law.

This formal boundary does not itself make a future prediction empirically true. It defines the minimum admissible interface every empirical branch must execute.

Engine receipt: `sha256 : a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086`.

15. Cross-claim synthesis

The root supplies occurrence. The One supplies exact self-wholeness. Positive count supplies complete finite generation trace. The part coordinate relates a nonempty held selection to its complete equal-fibre partition. Common refinement and exact pairing remove construction-dependent duplication without importing fraction arithmetic. The Fold supplies the first nontrivial equal returning distinction. Recursive extension supplies every finite held-label word. The form grammar closes every finite nested One/Fold assembly. Canonical enforcement supplies exact identity and comparison. The measurement law closes the outward interface to later observation.

These statements are compositional but not interchangeable. The One is not a number axiom. Count is not imported arithmetic. A held/whole coordinate is not yet an arithmetic field. The Fold is not an assumed bit. A Fold word is not an imported binary string. Canonical form equality is not an imported graph isomorphism algorithm. Measurement discipline is not evidence that an untested natural claim is correct.

This separation is scientifically important. Each downstream branch must derive its own operations rather than treating familiar correspondence as permission to import a model.

16. Machine verification and independent invalidation

The definitive command is:

```
python3 -m sft verify-all # macOS and most Linux systems
py -m sft verify-all      # standard Windows Python launcher
```

Installations exposing `python` may equivalently run `python -m sft verify-all`. These are host spellings of one logical command, `sft verify-all`; they execute the same verifier and do not create operating-system-specific scientific routes.

The completed foundation run reports:

- 96 unit, adverse-control and end-to-end tests passed;
- 1,264 of 1,264 core-engine executable lines observed;
- 15 core engine modules covered;
- ten admitted derivations freshly rerun in dependency order; and
- ten implementation-distinct validators freshly executed.

The coverage figure is an exact software execution statement, not a substitute for mathematical closure. Mathematical closure is carried by each claim's generator, census, decisions, minimality, named-shape proof and generality certificate. The test suite establishes that the shared engine executes its declared acceptance and rejection paths.

Independent invalidation requires no institutional permission. A reviewer can change a source, remove a candidate, add a survivor, alter a closure hash, fail a control or replace an independent validator. The engine returns the exact halted stage and preserves the rejected receipt. If a claimed complete grammar has an unrepresented structural alternative, presenting that alternative attacks the claim-specific completeness certificate directly.

17. Evidence custody and reproducibility

Each claim package contains registration, human WHY/DERIVATION/CHECK text, candidate census, elimination receipt, controls, certificate, executable source, independent validator and status. The authoritative census points only to model-admitted engine receipts. The ordered execution manifest must match the census exactly; omission or reordering fails complete verification.

Hashes bind source manifests, candidates, decisions, closures, controls, derivation seals, independent implementations, independent certificates and engine receipts. A hash does not prove scientific truth. It establishes which exact object was checked and prevents an altered object from silently inheriting the status of the checked one.

The V3 source does not depend on the prior-work ledger or conventional correspondence. Earlier SFT work may be compared only after a corresponding V3 claim seals. Agreement, stronger closure, weaker closure and disagreement are all preservable outcomes; prior answers cannot select the clean-room result.

18. Limitations and next boundary

Foundation closure does not mean that all mathematics or science has been derived. No arithmetic operation, algebra, probability law, automaton, complexity class, physical constant, quantum gate or biological relation is claimed in this paper. The part structure provides exact coordinates and equivalence witnesses, not yet the entire arithmetic of those coordinates.

All generality claims concern generated finite objects. The theory does not install a completed infinity or ungenerated continuum. Depth-independent means that a base/successor or structural-induction certificate covers every generated finite depth; it does not mean that an infinite object has been completed.

The next branch is mathematical foundations, beginning with exact arithmetic and number structure. It must cite these receipts but may not treat the desired conventional arithmetic as an answer table. Later empirical branches must use the one-way measurement law and the engine's capability and custody controls.

19. Open-science status, rights and participation

The platform is public, inspectable and reproducible with the Python standard library for its core verification route. Maria Smith retains copyright and scientific authorship. The paper and documentation are licensed CC BY 4.0; code is Apache-2.0. These permissions enable copying, inspection, modification and redistribution while preserving attribution. The Ernos Labs name is a separate conformance designation: use of that designation requires adherence to the published empirical and community standards.

Credentials cannot rescue a failed mechanical gate, and mechanical closure does not eliminate the need for domain review when natural observation is involved. Submissions, criticisms and independent invalidations are invited through Maria.Smith.Sftoe@gmail.com and <https://discord.gg/ucwGryVxGr>.

20. Conclusion

The V3 foundation closes a continuous operational chain from occurrence to the Fold and from the Fold to finite exact forms, while preserving a strict boundary against imported answer-producing models and target feedback. Ten claims, 2,450 generated alternatives and ten independent validators are bound into one reproducible census.

The foundation is not accepted because it resembles established mathematics, because software compiled, or because an authority endorsed it. It is accepted within its declared boundaries because every dependency is admitted, every candidate grammar is generated, every alternative is decided, one form survives, minimality and named-shape uniqueness close, adverse controls behave as registered and a distinct implementation recomputes the result. That is the standard the remainder of the knowledge tree must now meet.

Appendix A. Authoritative receipt identities

Claim	Engine receipt hash
Root	sha256:711864171e4d3a2f2734f0c2890965bcd81a0228349538751a3c80699c27d669
One	sha256:68332624c276dc5d0ddfd568a26b95e20d7d5665fcd7e825ff1b7b19bf1d51c
Count	sha256:4b2411716cbef2fb9fa7aac58b24ce5eb3c0c59fe7230fd703fd89b5260f6e23
Part	sha256:87b40affae50458ba5492af91778fbca79a8de0bc3b951e520535093c2b7b728
Fold	sha256:9e99f91d15e2b68a5c24f825093686863c168595319c2083e6a5b52b0314e003
Part equivalence	sha256:4f7c913f240527ec64d7688bf57fe95ae508f5445786dcbbdc2e6a7094476680
Fold assembly	sha256:8fa632be9e4048a5f2cf6749da4110e3a7b8b4d937051610dcce0583d7ea610d
Form grammar	sha256:b768689e7b061fdafa3a620befe79ddcee8bf5750cbbaf19f7b2a3156942be71
Form enforcement	sha256:84d18de18b1345bf506138493aebd7808bb0bd00cc3b318040e00751929c7c8a
Measurement boundary	sha256:a4e409f6bb8b203de746bd0c21d3f2566e8052003219490c657249e11f6f2086

Appendix B. File-level evidence route

For claim identifier <CLAIM>, reviewers should inspect:

```
claims/<CLAIM>/registration.json
claims/<CLAIM>/WHY_DERIVATION_CHECK.md
claims/<CLAIM>/candidate_census.json
claims/<CLAIM>/elimination_receipt.json
claims/<CLAIM>/controls.json
claims/<CLAIM>/certificate.json
claims/<CLAIM>/execution.py
claims/<CLAIM>/independent_validator.py
receipts/engine/model_admitted/<CLAIM>-<receipt-prefix>.json
```

The executable scientific modules are under `sft/foundation/`. The authoritative order is `census/execution_manifest.json`; the admitted projection is `census/claims.json`; the frozen scope is `publications/inventories/foundation.json`. The accompanying evidence map binds every paper section to those exact files and hashes.

References

1. Smith M. *There Is No Nothing: A Premise-Free Operational Foundation and an Open Verification Platform for Smithian Fold Theory*. Ernos Labs Methods and Foundation Paper 001. 2026. doi:10.5281/zenodo.21514890.
2. Smith M. *Smithian Fold Theory Scientific Constitution*. V3 clean-room repository, `CONSTITUTION.md`, 2026.
3. Ernos Labs. *The Single SFT Admission Engine*. V3 clean-room repository, `docs/ENGINE_AUTHORITY.md`, 2026.
4. Ernos Labs. *Comprehensive Branch-Paper Protocol*. V3 clean-room repository, `publications/BRANCH_PAPER_PROTOCOL.md`, 2026.